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Special Collection:
Integrating In Situ, Remote Sensing, And Physically Based Modeling Approaches to Understand Global Freshwater Ice Dynamics

# One-Hundred Fundamental, Open Questions to Integrate Methodological Approaches in Lake Ice Research

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**Key Points:**

- We formulated 100 questions to progress primary research methods of lake ice research: in situ, satellite remote sensing, and modeling
- We described avenues to integrate the primary methods and incorporate concepts from adjacent fields to expand lake ice research
- Pragmatic next steps include grassroots efforts to create communities of practice focused on skill development and knowledge sharing

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## Abstract

The rate of technological innovation within aquatic sciences outpaces the collective ability of individual scientists within the field to make appropriate use of those technologies. The process of in situ lake sampling remains the primary choice to comprehensively understand an aquatic ecosystem at local scales; however, the impact of climate change on lakes necessitates the rapid advancement of understanding and the incorporation of lakes on both landscape and global scales. Three fields driving innovation within winter limnology that we address here are autonomous real-time in situ monitoring, remote sensing, and modeling. The recent progress in low-power in situ sensing and data telemetry allows continuous tracing of under-ice processes in selected lakes as well as the development of global lake observational networks. Remote sensing offers consistent monitoring of numerous systems, allowing limnologists to ask certain questions across large scales. Models are advancing and historically come in different types (process-based or statistical data-driven), with the recent technological advancements and integration of machine learning and hybrid process-based/statistical models. Lake ice modeling enhances our understanding of lake dynamics and allows for projections under future climate warming scenarios. To encourage the merging of technological innovation within limnological research of the less-studied winter period, we have accumulated both essential details on the history and uses of contemporary sampling, remote sensing, and modeling techniques. We crafted 100 questions in the field of winter limnology that aim to facilitate the cross-pollination of intensive and extensive modes of study to broaden knowledge of the winter period.

## Plain Language Summary

Limnology is a field that has largely been relegated to the summer period when the water is open. Seasonal ice cover limits access and presents numerous challenges to studying lakes, limiting the understanding of seasonal lake processes. Research within the last decade has dramatically increased on lake ice cover, primarily through in situ observation, satellite remote sensing, and modeling. However, blending these methods of inquiry synergistically is an outstanding challenge. To facilitate forthcoming lake ice research, we present a set of 100 questions that deal individually with each research method and best practices in how to integrate these methods. These questions can be used by the lake ice community to drive research within a quickly growing field and to help round out winter as a seasonal aspect of limnology more broadly rather than a subdiscipline.

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## 1. Introduction

Ecosystem services provided by lake ice generate economic, recreational, cultural, and environmental value (Knoll et al., 2019; Leppäranta, 2023a; Magnuson & Lathrop, 2014). Economically, ice facilitates transportation and trade through ice roads in northern and remote regions where air travel is cost prohibitive (Hori et al., 2017; Newton & Mullan, 2021; Woolway et al., 2022). Frozen lakes create opportunities for ice-related activities, such as ice fishing, which attracts winter enthusiasts and generates income for local communities (Knoll et al., 2019; Orru et al., 2014). Recreationally, lake ice offers activities such as skiing, ice skating, and ice hockey, promoting physical well-being and community engagement in the cold and dark winter months but also includes a safety issue (Magnuson & Lathrop, 2014). Lake ice also holds cultural value with multi-generational traditions that include trade routes and fishing since the prehistoric era (Leppäranta, 2023a) and more recently monitoring ice conditions (Sharma et al., 2022). Understanding the importance of lake ice helps us recognize its role in water conservation, ecosystem health, economic growth, recreational enjoyment, and cultural heritage.

Ice regulates the water cycle by isolating lakes from the atmosphere, by reducing evaporation, and storing precipitation in snow cover. Ice also functions as a barrier between the water column and the atmosphere, resulting in a relatively constant thermal structure under the ice layer and generally damping turbulent mixing (Kirillin et al., 2012). Ample biological and chemical activity supports biodiversity under ice (Powers et al., 2017), including higher than anticipated primary production, dynamic phytoplankton and zooplankton communities, niche partitioning in fish species, and timing disruption of biological events and food webs in freshwater ecosystems (Bramburger et al., 2022; Hampton, Galloway, et al., 2017; Hébert et al., 2021; McMeans et al., 2020). Furthermore, lake ice cover and cold water temperatures influence nutrient cycling and biogeochemical cycles (Jensen, 2019; Swann et al., 2020). Ice-covered periods can represent more than a quarter of the total $CO_2$ and $CH_4$ budget (Denfeld et al., 2018). Under-ice dissolved oxygen dynamics can impact ecosystems through consistent oxygen depletion (Jane et al., 2021; Smits et al., 2021) and nitrogen accumulation through ammonium oxidizing microbes (Powers et al., 2017). These chemical interactions under ice are further complicated when considering system size and near-shore and off-shore dynamics in large systems, such as the Laurentian Great Lakes (Ozersky et al., 2021).

Widespread, unprecedented lake ice loss is being recorded in response to global warming (Huang et al., 2022). Ice formation also plays a distinct role in the lake-atmosphere heat balance due to the albedo effect and loss of ice in larger water bodies, which can, in turn, affect the climate (Prowse et al., 2011). Studies on freshwater lake ice responses to climate change have shown that rising temperatures result in shorter ice cover duration, thinner ice, and reduced ice cover extent in lakes globally (Grant et al., 2021; Huang et al., 2022; Li et al., 2022; Magnuson et al., 2000; Sharma et al., 2019). Changes in the timing of the ice cover season have been shown to change the mixing regime in some lakes from meromixis to dimixis, with hypolimnetic increases in dissolved oxygen (Flaim et al., 2020). Detecting and predicting lake ice responses to climate change involves in situ field observations, remote sensing, and modeling, which provide valuable insights into vulnerability and the need for climate change mitigation and adaptation strategies. In situ field measurements include observations of ice phenology (timing of ice-on and ice-off), ice thickness, snow thickness, and ice quality (i.e., the ratio of black and white ice within an ice column). In situ measurements have been collected for centuries and some precede the Industrial Revolution (Livingstone, 1997; Magnuson et al., 2000). These records demonstrate how anthropogenic-driven climate change since the Industrial Revolution has affected lake ice (Fujiwhara, 1921; Sharma et al., 2016). Satellite remote sensing provides data on ice cover extent and phenology (Murfitt & Duguay, 2021; Wynne & Lillesand, 1993) and ice thickness (Kang et al., 2014; Kheyrollah Pour et al., 2017; Mangilli et al., 2022, 2024), allowing for repeated, large-scale monitoring over time. Satellite observations have provided valuable insights into the spatial heterogeneity of patterns in ice cover and loss (Karetnikov et al., 2017; S. Zhang et al., 2021), which exhibit varying degrees of sensitivity and response to warming temperatures (Li et al., 2022; X. Wang et al., 2022). Additionally, advanced technologies, such as aerial drones and underwater sensors (Brown & Duguay, 2011a; Pierson et al., 2011) provide detailed information on ice cover properties (Alfredsen et al., 2018; Lally et al., 2019; Pierson et al., 2011). Climate models coupled with lake models support assessment of the potential impacts of future shifts in climate on lake ice and can be used to evaluate the influence of anthropogenic climate change (Brown & Duguay, 2011b; Grant et al., 2021; Huang et al., 2022). Integrating monitoring and modeling approaches enhances our understanding of lake ice dynamics and aids in identifying trends and patterns related to climate change (Sharma et al., 2020).

The power of integrating data collection and modeling methods of ice phenology is derived from the ability to compensate for the limitations of a given data collection method or model. For example, modeling work has long relied on both in situ and remote sensing observations to validate results (Duguay et al., 2003; Ménard et al., 2002; Vavrus et al., 1996). Similarly, remote sensing relies, at least initially, on in situ observation networks to verify optically and microwave-derived ice phenology (Cai et al., 2022; Zhang & Pavelsky, 2019; S. Zhang et al., 2021) or thickness (Li et al., 2022; Mangilli et al., 2022). In situ data collection captures observations of individual lakes intensively, but scaling observation networks, even regionally, can be logistically challenging, expensive, and potentially dangerous (Block et al., 2019). Even within a single lake, data collection methods can differ between sites, resulting in the discrepancies in the percentage of ice cover that constitutes the date of ice-on (e.g., 80% vs. 100% surface area coverage). The continuity of long-term data sets should be preserved, but one of the benefits of merging data collection and modeling methods is creating a comprehensive lexicon that can apply across ecosystems (Sharma et al., 2020). Other challenges to integration come from blending technologies within a discipline to be exported for use in other disciplines. For example, remote sensing can involve a broad swath of sensors that have varying spatial and temporal resolutions. Future work that uses remote sensing data may benefit from blending two or more complementary sensors to offer more complete visualization of lake ice phenology on one or many lakes (Ghassemian, 2016; X. Wang et al., 2021). Therefore, blending methods of data collection enhances our understanding of single systems and landscape-scale analyses.

In this perspective, we—a group of lake ice researchers—describe the three primary methods of data collection and analysis of lake ice: in situ, remote sensing, and modeling. We offer a brief history of each topic and their usefulness for lake ice study, along with a collection of open questions and challenges that have yet to be comprehensively answered. Interactions between data collection methods are captured in Figure 1 of Sharma et al. (2020), which incentivized this group of ice researchers to collaborate on enumerating open questions regarding these research methods and the integration of the subsequent data. This group developed these questions through community discussion and pruned similar questions and questions already answered within the available literature. We then discuss the integration of those methods. We also describe possible future directions of research with a collection of open questions that make use of these different data collection and analysis methods. While many of these proposed questions could be addressed by incorporating in situ, remote sensing, and modeling methods individually or in combination, we offer these broad groupings of frontiers to suggest the most tractable paths forward for lake ice research.

## 2. In Situ Observations

The most common lake ice data collection methods are human observations of lake ice phenology (i.e., the timing of ice-on and ice-off events) (Sharma et al., 2024). Such in situ observations began decades ago, often because of curiosity but also to keep track of ice conditions for recreation and traffic on lakes (Sharma et al., 2024). Over time, ice records from individual lakes became available, and the first studies on long-term trends in ice phenology were published (Assel & Robertson, 1995; Livingstone, 1997; Magnuson, 2021; Robertson et al., 1992). Subsequently, researchers moved to studying ice phenology records within a region or a country to obtain a more expansive perspective on how lake ice is changing across lakes within a landscape (W. L. Anderson et al., 1996; Duguay et al., 2006; Hodgkins et al., 2002; Korhonen, 2006; Palecki & Barry, 1986).

A breakthrough occurred in 1996 during a workshop held in Wisconsin, USA. A group of lake ice researchers from around the world gathered to aggregate long-term ice data sets and ask questions about how lake ice is changing around the Northern Hemisphere (Magnuson et al., 2000). The group aggregated the data sets and built the first global trend analysis of lake ice phenology, which has become the rubric by which to study lake ice (Benson et al., 2012; Brown & Duguay, 2010; Duguay et al., 2006; Magnuson et al., 2000; Sharma et al., 2022). This study elucidated the overall loss of lake and river ice through later ice formation and earlier ice breakup in response to warming air temperatures (Magnuson et al., 2000). Over a 150-year period from 1846 to 1995, they observed that ice-on was 5.8 days per century later and ice-off was 6.5 days per century earlier (Magnuson et al., 2000). Afterward, Benson et al. (2012) expanded on the earlier study by updating the records to 2004 and adding lakes with 30, 100, and 150 years of data. Benson et al. (2012) not only calculated trends in lake ice records, but also assessed variability and extreme events, defined as extremely early or late ice cover or years without annual ice cover for lakes that normally freeze. Sharma et al. (2021) further updated the lake ice data to 2019 for approximately 70 lakes and built a time series of up to 204 years of data. They calculated trends in ice-on as 11 days later and ice-off as 6.8 days earlier over the entire time-series. Notably, in the last 25 years, lake ice loss

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has accelerated, on average six times faster around the Northern Hemisphere, but with large geographical differences owing to the nonlinear responses of lake ice breakup to global warming and lake geomorphology (Weyhenmeyer et al., 2005, 2011). Although we have a good understanding of long-term trends in lake ice loss, we still have remaining questions on what is causing interannual and interdecadal variability and their sensitivity to air temperature.

Recent progress in the autonomous methods of data collection, storage, and transfer provided a new dimension to in situ lake ice observations. Following advances in oceanographic ice observations (Jackson et al., 2013; Polashenski et al., 2011; Richter-Menge et al., 2006), lake ice studies started to actively involve autonomous monitoring of ice properties at previously unavailable temporal and spatial resolutions (Aslamov et al., 2021; Brown & Duguay, 2011a; Y. Cheng et al., 2020; B. Cheng et al., 2021; C. Wang et al., 2005). Use of modern telemetry and sensing technologies with low power consumption has allowed real-time registration of sub-daily to seasonal variations of ice thickness and temperature with possible expansions that can include ice structure and optical properties. Compared to phenology records from individual observers, the new methods provide insight into physical mechanisms behind ice phenology to bolster ice duration trend observations. Now in their early stages, autonomous lake ice monitoring networks may quickly develop into major sources of information about the dynamics of ice cover in remote locations akin to the development of weather monitoring networks in the 1900s and the gliders and buoy ocean monitoring networks in the early 2000s (Fiebrich, 2009; Schwing, 2023). Similar to the advances in climatological and oceanic studies, autonomous in situ ice monitoring can integrate into a multi-approach framework, merged with the comprehensiveness of field experiments, global spatial coverage of remote sensing, and deterministic skills of modeling (Figure 1).

It is clear that lake ice loss is driven by warming air temperatures in response to anthropogenic-driven climate change (Grant et al., 2021) alongside multi-annual or multi-decadal climate oscillations (e.g., El Niño Southern Oscillation, North Atlantic Oscillation, Pacific Decadal Oscillation, and Quasi-Biennial Oscillation) impact the amplitude or frequency of those primary climatic drivers (Bai et al., 2012; Bonsal et al., 2006; Ghanbari et al., 2009; Robertson et al., 2000; C. R. Walsh & Patterson, 2022). However, studies that describe temporal and spatial patterns of ice loss and potential ice loss patterns under future climate change (Brown & Duguay, 2011b; Sharma et al., 2019; S. E. Walsh et al., 1998; Weyhenmeyer et al., 2011) need to be built upon to capture the rapid pace (Imrit & Sharma, 2021) and high variability (Richardson et al., 2024) of lake ice loss. Increasingly, lakes are experiencing freeze-thaw events within a season in response to fluctuating air temperatures and rain-on-snow events (Sharma et al., 2021). These same forces drive a shift in the quality of lake ice, where warmer temperatures and precipitation phase changes lead to more white ice cover (Brown & Duguay, 2011b; Weyhenmeyer et al., 2022). However, no global studies exist to understand the drivers or impacts of freeze-thaw events on ice quality, as we do not yet have a homogenous, high-resolution, global data set with comparable ice observations and potential drivers. Given the nature of in situ observations, seasonal, iterative freeze-thaw events complicate how ice phenology is defined (e.g., which formation and breakup dates are recorded during a season). Meanwhile, ice quality lacks a single cohesive definition and data paucity limits the detection of long-term changes (Korhonen, 2006; Weyhenmeyer et al., 2022). These rapid changes in climate and their influence on lake ice must also be understood on a broad spatial scale, given the continental or global heterogeneity of lake ice loss recorded from long-term data (Figure 1) (Newton & Mullan, 2021).

## 3. What Are the Remaining Questions and Challenges for In Situ Research?

### 3.1. Methodology, Data Quality, and Data Harmonization

1. What is the best method to harmonize local and subjective ice observations globally so that they are interoperable through standard operational protocols (SOP), and how have changing definitions of complete ice cover through time affected the estimated trends?
2. How can ice quality be defined, using a shared approach to establish SOPs?
3. How can enhanced standards for in situ monitoring ensure the acquisition of high-quality data for forecasting ice-on, ice-off, ice duration, and ice thickness under climate change scenarios?
4. Can common SOPs for lake ice data and under-ice ecology permit interoperable data to be incorporated within existing repositories?
5. Can SOPs be created to understand under-ice dynamics, considering changes to ice phenology, thickness, and quality?

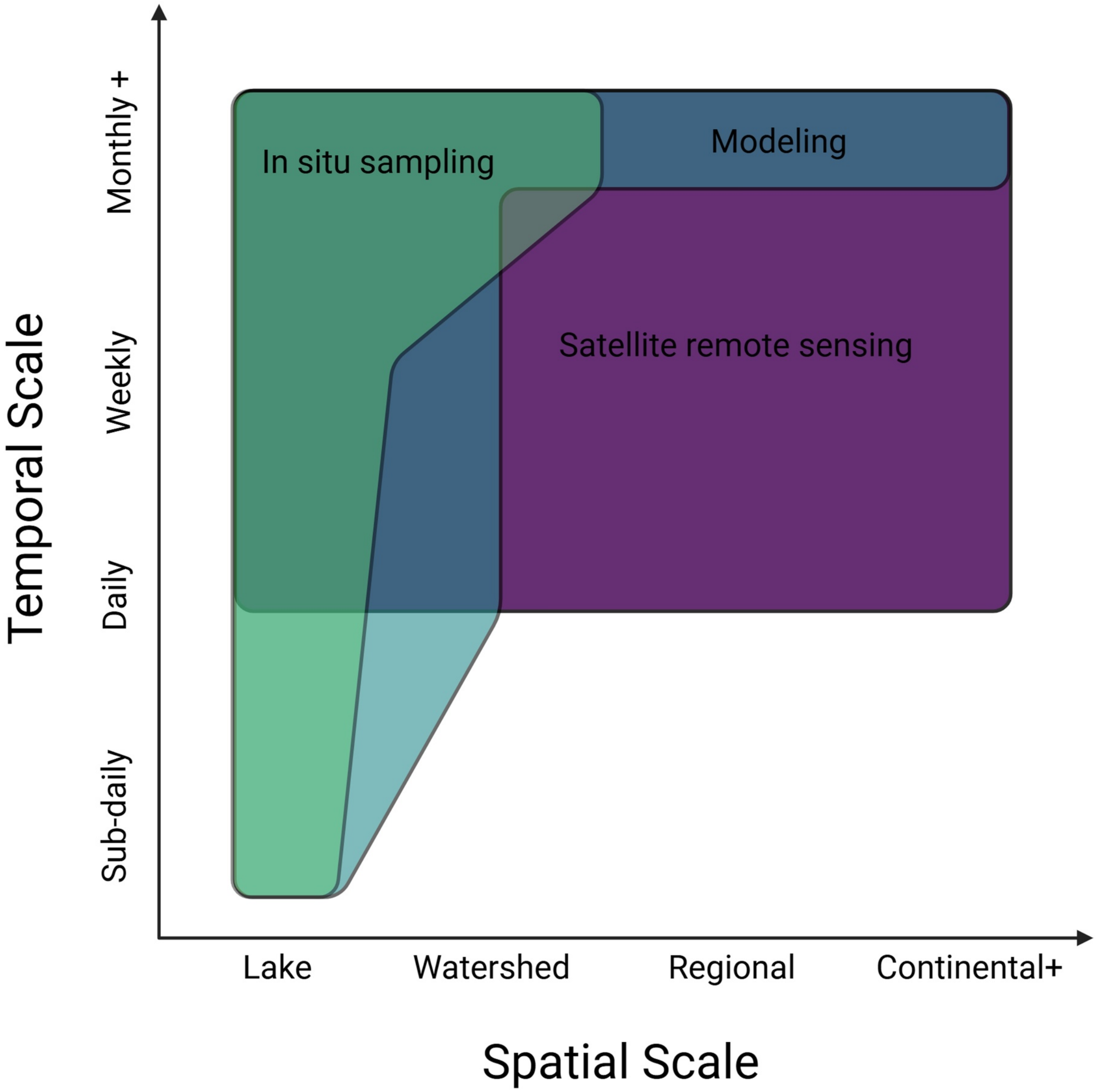


**Figure 1.** Extent and limitations of data acquisition methods. The possible spatial and temporal scales of different data collection methods are depicted. Temporal scale varies between sub-daily (e.g., multiple sampling campaigns within a lake per day to track diel variation) to monthly and greater (e.g., yearly sampling campaign to a less accessible lake). Spatial scale varies between single-lake sampling and continental and greater (i.e., lakes from various watersheds within a continent or a global lake study from international collaborators). Created in BioRender by Culpepper, BioRender.com/k45h898.

6. Can observation methods and SOPs within sea ice research be incorporated into lake ice research and under-ice biogeochemistry?
7. What are the major new features that in situ monitoring should focus on to complement and calibrate remote sensing observations?
8. We have biases in sampling because of unsafe ice, so can we develop new techniques to safely measure ice during shoulder seasons and freeze-thaw events?
9. How can effective monitoring networks be designed, and which lakes are prime candidates for establishing continuous autonomous ice monitoring?
10. How can in situ data be integrated within real-time management models?
11. How can community science best augment ice monitoring (e.g., by maintaining autonomous stations or performing auxiliary observations while adhering to SOPs)?

### 3.2. Understanding Lake Ice Dynamics in Single Lakes

12. How are ice phenology, ice thickness, and ice stability impacted by atmospheric and climatic oscillations, such as increasing Arctic polar vortex events?
13. To what extent does increased, decreased, and/or variable salinity impact ice cover dynamics?
14. What is the heterogeneity of ice growth and ice cover within a lake?
15. What is the relationship between ice phenology, seasonal mean and maximum thickness, and ratios of black and white ice?
16. What is the biochemical composition of ice and how is it expected to change?
17. How do lake ice phenology, thickness, and quality impact water quality in the following open-water season?

### 3.3. Role of Climatic Factors and Lake Geomorphology on Ice Cover

18. How does lake morphometry affect the quality of ice?
19. How do watershed and landscape characteristics (e.g., land cover, topography, land use, hydrologic connectivity, etc.) influence lake ice phenology, mean and maximum ice thickness, and the ratio of black and white ice?
20. How does ice quality vary in time and space and how can it be predicted?
21. How does trophic status and water quality in the open-water season affect lake ice quality and lake ice composition?

### 3.4. Forecasting Lake Ice in a Changing Climate

22. How will changing ice conditions be defined, incorporated, and analyzed with respect to intermittent ice cover, freeze-thaw events, and winter rain events?
23. How will the bearing capacity of ice be affected by climate change, considering declining ice thickness and changing ratios of black and white ice?

## 4. Satellite Remote Sensing

Remote sensing offers an opportunity to monitor freshwater hydrology (lakes and rivers) and measure key parameters without the requirement of physically sampling the system (Figure 1). Consistent spaceborne acquisitions allow for the observation of small and large lakes that exist in remote regions where field observations would be logistically cumbersome. The observation of lake properties during the open water (water extent, quality, persistence) and frozen (ice phenology, extent, type, thickness) seasons have taken place since the 1970s with airborne radars (Elachi & Apel, 1976; Weeks et al., 1978), spaceborne radiometers (Cai et al., 2022; Hall et al., 1981; Kang et al., 2010; Su et al., 2021), and optical sensors onboard Landsat and Moderate Resolution Imaging Spectroradiometer (MODIS) missions, with consistent records beginning in the 1980s and 2000s, respectively (Q. Yang et al., 2023). Retrieval of lake variables from remote sensing, including those listed on the Global Climate Observing System's Essential Climate Variables (e.g., lake ice thickness, percentage of ice cover, etc.) (GCOS, 2021), has largely been intermittent and opportunistic and is often the result of research collaboration with national and international space agencies and government divisions to access third-party mission data. While satellite mission concepts such as CoREH2O have been proposed to observe and monitor terrestrial hydrology (Rott et al., 2008), the Surface Water and Ocean Topography (SWOT) mission was launched in 2022 with the goal of increasing knowledge of lake, river, and ocean physical properties (Biancamaria et al., 2016). Nonetheless, the utility of optical, thermal, and microwave technology (passive, imaging radar, and radar altimetry) has repeatedly demonstrated effective retrievals of thermodynamic lake status and quantifications of physical features of the ice column (Figure 2) (Brown & Duguay, 2012; Murfitt et al., 2022).

Past and current satellite remote sensing platforms have contributed to advances in the retrieval of dates associated with lake ice phenology, ice cover extent, and physical parameters, such as ice thickness, using a variety of sensors, technologies, algorithms, and machine-learning techniques. Remotely sensed monitoring of lake surface water temperature (LSWT) became prevalent with the launch of optical and thermal satellite sensors that imaged at similar spatial resolutions (500 m– 1 km), namely, MODIS (Oesch et al., 2005; Pour et al., 2014), the Advanced Along-Track Scanning Radiometer (AATSR) (Fichot et al., 2019; Layden et al., 2015; MacCallum & Merchant, 2012) and the Advanced Very High Resolution Radiometer (AVHRR) (Lieberherr & Wunderle, 2018; B. Liu et al., 2019; Oesch et al., 2005; Schwab et al., 1999). Advancements in cloud computing have resulted in the

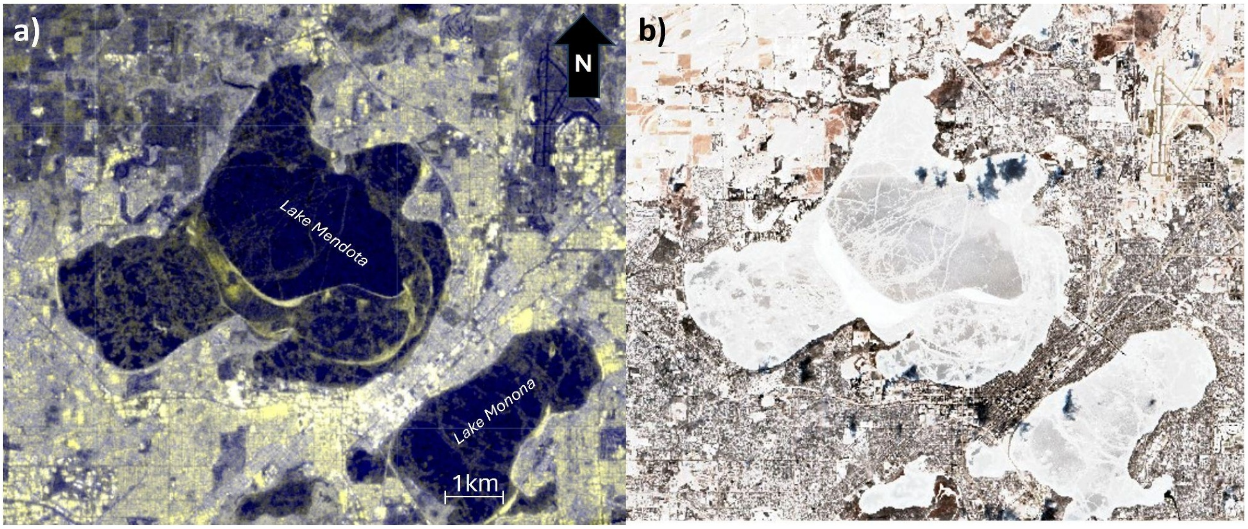


**Figure 2.** Lake ice visualizations from active and passive sensors. Microwave (Sentinel-1) false color composite (R = VV, G = VV, B = VH) and true color composite (Sentinel-2) of Lake Mendota and Lake Monona, Wisconsin, USA on 8 February 2019. The relatively thin snowpack during this period in 2019 allows for both the viewing of ice dynamics (consolidated thin pan in the main lake basin) to be observed by both microwave and optical imagery. Sentinel-1 and Sentinel-2 accessed via the European Space Agency.

development of global LSWT products (Cook et al., 2014) accessible through Google Earth Engine (Herrick et al., 2023). These satellites and LSWT products have also been used to develop ice cover and thickness products, such as and the LakeTEMP data set, which provides global LSWT and ice cover for over 1.4 million lakes (greater than 0.1 km$^2$) from Landsat 8 thermal radiance observations (Korver et al., 2024).

Similar developments for optical retrieval of lake ice phenology have occurred in the last 30 years, whereby cloud computing and online repositories are being utilized to retrieve hemispherical or global retrievals at high spatial resolution. The MODIS Snow and Ice global mapping project was a significant advancement in the detection of ice covered pixels using surface reflectance thresholds (Gou et al., 2017; Hall et al., 2001; Qi et al., 2019; S. Zhang & Pavelsky, 2019), with improvements to phenological retrievals developed using machine learning classifiers (Tom et al., 2018; Wu et al., 2021; X. Yang et al., 2021). The random forest classifier of Wu et al. (2021) has been implemented in the processing chain of the global lake ice cover (LIC) product generation as part of ESA's CCI project. The LIC product is delivered on the same grid as the LSWT product mentioned above, along with water-leaving reflectance, water level and extent for 2024 lakes (Carrea et al., 2023), amongst the largest in the world, allowing for multivariate analyses of the response of lakes to climate change in the last two to three decades.

Researchers have successfully retrieved ice phenology using both passive microwave (PM) and active microwave (AM) satellite sensors; however, drawbacks persist with each technology. PM observations at various frequencies (6.9, 19, and 37 GHz) can detect the presence of lake ice (Cai et al., 2022; Du et al., 2017; Kang et al., 2012), with emission directly correlated to ice thickness for observations on a sub-daily basis (Kang et al., 2014). However, the relatively low strength of emission from the surface results in the requirement of the sensor to incorporate a larger instantaneous field of view (i.e., the angular field of view of a sensor, independent of height), with nominal pixel sizes ranging from 3.125 to 25 km, depending on the wavelength and processing level (X. Wang et al., 2022). The coarse spatial resolution of PM sensors restricts ice phenology products to lakes greater than 50 km$^2$. For global context, there are approximately 713,000 lakes located in the Arctic circle (>60°N), roughly 712,500 of which are smaller than 50 km$^2$. Nearly all small-to-medium sized lakes are unable to be observed by PM without mixed pixel effects (Messager et al., 2016). Retrieving ice phenology for smaller lakes has repeatedly been demonstrated using AM (Geldsetzer et al., 2010; Kouraev et al., 2008; Murfitt et al., 2018; Surdu et al., 2015); however, longer revisit times (i.e., time period between images of the same location) from 11 to 26 days generally increases the root mean square error (RMSE) compared to that of PM, with the exception of high latitude polar regions where swaths overlap, effectively reducing revisit times to nearly daily (Murfitt & Duguay, 2020). Active microwave observations have also been useful in the retrieval of other lake ice properties, including the timing of ice freezing to the bottom of shallow lakes (i.e., bedfast ice) (Antonova et al., 2016; Engram et al., 2018; Surdu et al., 2015), areas of increased methane ebullition (Engram et al., 2020; Pointner et al., 2021), and ice thickness retrieval (Beckers et al., 2017; Duguay & Lafleur, 2003; Gunn et al., 2015; Mangilli et al., 2022, 2024). In a recent investigation, Ghiasi et al. (2024) have also demonstrated the potential of global navigation satellite system reflectometry (GNSS-R) for the monitoring of ice phenology on large lakes.

Machine learning approaches can be used to process and interpret image data, although generalization across space and time is challenging. Supervised machine learning techniques (e.g., random forest models) require ample training data for discrete image labeling, which requires collaboration between lake ice researchers and remote sensing experts to develop a large, labeled data set for training. Unsupervised learning techniques have been successfully implemented in radar imagery (Scott et al., 2020) and optical data (Heinilä et al., 2021) for the classification of open water and ice pixels. Additionally, recent research has incorporated radar altimetry (Mugunthan et al., 2023) and SAR imagery (Hoekstra et al., 2020; Ma et al., 2021) along with multiple machine learning methods to classify ice cover and its phenology (e.g., ice formation, growth, melt onset, and breakup). Unsupervised approaches (e.g., k-means clustering) may allow for characterization of similar "types" of ice within an image, but without in situ validation, the typology may not be meaningful to lake ice researchers. The true pinch point of machine learning applications across many fields of study (even beyond winter limnology) may be the requirement of robust train-validate-test splits, which require large amounts of data. More intentional model development will allow for greater generalizability across new and/or unseen regions or winters through non-random, thoughtful train-validate-test assignment that considers spatial and temporal autocorrelation. Continued work to create such models is especially important as it becomes easier for researchers to use machine learning techniques.

Recent advancements in remote sensing technology and processing techniques continue improving the understanding of electromagnetic interaction with water or ice to understand and distinguish between the two media. The development of polarimetric decomposition (Atwood et al., 2015; Gunn et al., 2018), radiative transfer modeling (Murfitt et al., 2023), and machine learning (Wu et al., 2021) has set the stage for a significant increase in the accuracy and the frequency of lake ice property retrieval through the next phase of satellite sensors (e.g., Surface Water and Ocean Topography (SWOT) launched in 2022, and the NASA-ISRO Synthetic Aperture Radar (NISAR) to be launched in 2025). Many questions put forth by remote sensing experts focus on how new sensors will improve retrievals based on two factors: (a) the improvement in the technology itself, and (b) shorter revisit time offered by constellation missions (i.e., groups of interoperable satellites). Conversely, many of the questions also focus on overcoming challenges for lake properties that remain difficult to retrieve or lake processes that are too dynamic to be captured from space. For example, snow depth or snow water equivalent atop ice is currently difficult to resolve outside of case studies using uncrewed autonomous vehicle technology (Gunn et al., 2021), and is not currently resolved from spaceborne sensors. With these challenges and knowledge deficits in mind, we have collated the following questions that are critical to the remote sensing of lake ice.

## 5. What Are the Remaining Challenges and Big Questions for Remote Sensing Lake Ice Research?

### 5.1. Detection of Physical, Ecological, and Biogeochemical Ice Characteristics

24. Can a single time series of long-term ice cover be created with remote sensing (a few decades for millions of lakes)?
25. How can remote sensing be used to observe dynamic, short-term events like intra-annual freeze-thaw cycles?
26. To what extent can remote sensing be used to distinguish between different ice types (black ice, white ice, snow on ice, brackish ice)?
27. What methods can be used to retrieve physical properties of snow on ice (e.g., snow depth, snow density, water equivalent)?
28. How can surface roughness (i.e., ice thickness irregularity across the ice layer) of the ice-water interface be measured? Fine-scale (mm to cm) measurements of the ice bottom surface roughness currently cannot be captured but are important for understanding microwave interaction and ice growth dynamics.
29. What role does surface roughness play at the ice-water interface with respect to how backscatter increases during ice growth?
30. Can remote sensing be used to quantify ice algae in the snow pack with optical acquisitions?

31. How do impurities, such as silt, dirt, or algae, affect snow albedo, and can impurities be resolved using remote sensing to determine the impact on the ice breakup season?
32. Can multiple satellite altimeters with different working bands (e.g., C, Ku, Ka, and laser) be combined to differentiate snow depth and ice thickness?
33. Can remote sensing be utilized to observe and quantify the size of cracks and pressure ridges to anticipate break-up dynamics?

### 5.2. Remote Sensing Methods and Technologies

34. Will interferometry (i.e., surface deformation detected from overlapping images or repeat passes) be a potential solution to retrieve ice thickness directly from synthetic aperture radar (SAR) acquisitions?
35. How can more frequent thermal observations coupled with SAR be used to retrieve ice properties during the freeze-up season and polar night?
36. At what level of error/uncertainty is remote sensing no longer useful for answering questions about lake ice cover?
37. Regardless of latitude or observation technology utilized, it is difficult to quantify ice-on and/or freeze-up relative to open water. What combination of satellites will improve these estimates?
38. What will the role of GNSS-R active/passive remote sensing be in monitoring lake ice phenology and thickness in the wake of GNSS spaceborne sensor development?

### 5.3. Applications Across Scales/Methods

39. How can technology (e.g., better in situ sensors) or models overcome the lack of in situ measurements spatially and temporally coincident with satellite acquisitions?
40. How effectively can data assimilation of optical and radar sensors coupled with lake ice models fill the gap in remote sensing acquisitions?
41. How can machine learning be best applied to remote sensing data for ice property retrieval outside of ice presence/absence detection?
42. Can we build a common language and/or understanding between remote sensing and limnology/aquatic ecology of the questions, terms, and overarching goals?
43. How well do machine learning/deep learning methods generalize retrievals of lake ice from satellite sensors across space and time?

## 6. Modeling

Lake ice models have been used to predict ice cover dynamics (Duguay et al., 2003; Leppäranta, 1983, 2023b), develop connections between and amongst drivers of ice dynamics (Leppäranta & Wang, 2008), and project long-term changes to ice dynamics under future warming climate scenarios (Brown & Duguay, 2011b; Li et al., 2022). Historically, in alignment with the closely related field of water temperature modeling, two primary model architectures have been used: statistical and process-based models (Piccolroaz et al., 2024). Studies progressed from single-lake statistical or processed-based models to expand both to regional (Robertson et al., 1992; Sharma et al., 2019, 2021; Vavrus et al., 1996; S. G. Williams & Stefan, 2006) and then global scales (Huang et al., 2022; Walsh et al., 1998; X. Wang et al., 2022; Woolway et al., 2021) (Figure 1).

Lake ice modeling began with statistically based models used to describe relationships between meteorological variables (e.g., air temperature) and lake characteristics (e.g., elevation, depth, etc.) to assess both lake ice phenology (ice on, ice off, duration) (Robertson et al., 1992) and thickness (Leppäranta, 1983). Statistical models are data-driven models that use observational data to establish relationships between lake ice phenology and environmental drivers (Karetnikov et al., 2017; Leppäranta & Wen, 2022). Models advanced to use combinations of lake morphological variables (e.g., mean lake depth) as a proxy for heat storage capacity alongside high-frequency meteorological inputs (e.g., air temperature, precipitation, and solar radiation) to detect lake ice phenology with high accuracy across a range of lake sizes (Vavrus et al., 1996). Statistically based models confirm that daily mean air temperature is the primary driver of lake ice phenology (Sharma et al., 2019) and thickness, while latitude and elevation are also influential determinants of ice phenology. Lake-specific characteristics, like shoreline complexity and depth also play a role. These modeling approaches have also been used

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to forecast lake-ice cover under future climate scenarios across local (Robertson et al., 1992) and large spatial scales (Filazzola et al., 2020; Sharma et al., 2019, 2021).

Process-based models incorporate mathematical descriptions of the physical principles governing lake ice dynamics (Duguay et al., 2003; Stepanenko et al., 2019; Toffolon et al., 2021; Y. Yang et al., 2012). Most process-based lake ice models are based on a one-dimensional framework, which represents the lake as a vertical profile in space, but increasingly more studies use two-dimensional and three-dimensional models (E. J. Anderson et al., 2018; Cannon et al., 2023; Fujisaki-Manome et al., 2020; F. Wang et al., 2006; Xue et al., 2022). Low-dimensional, process-based models that assume the lake volume as a box or continuum are also commonly applied to explore lake ice phenology (Flaim et al., 2020; Piccolroaz et al., 2021). Early process-based models were built using heat conduction to expand temperature models from the open-water seasons to the winter season (Gu & Stefan, 1990). Models attempting to approximate physical processes such as evaporation included ice formation when surface water temperatures decreased below 1°C by incorporating the latent heat of fusion into the heat balance at the surface (Hostetler & Bartlein, 1990). Snow cover is needed as an interactive component in a process-based lake ice model to produce realistic ice thickness (and quality) and ice-off, as well as under-water light conditions. Specific size thresholds (e.g., shallow lakes in the Arctic) in conjunction with thermodynamic modeling approaches borrowed from sea ice modeling gave way to outputs that capture the complex interactions of heat loss, precipitation, and ice formation, even capturing ice quality (black ice and white ice) (Duguay et al., 2003). Contemporary process-based models (Hipsey et al., 2019) have been thoroughly verified on a few individual lakes where the mechanisms of ice phenology are well understood from decades of intensive sampling. The validation from intensive sampling enabled the modeling of challenging case studies, such as irregularly freezing system (Bueche et al., 2017). Process-based models have been used to hindcast lake ice conditions to fill in gaps where in situ or remotely sensed data are unavailable (Bernhardt et al., 2012; Grant et al., 2021; Magee et al., 2016) or to recreate ice dynamics of lakes to understand important ecological dynamics of past decades. Ice-off relates to internal melting and mechanical deterioration, where final disappearance of ice is rapid (Y. Zhang et al., 2023). Process-based models have allowed scientists to assess impacts of wind speed (Magee & Wu, 2017), snow cover (Duguay et al., 2003), and precipitation (Fujisaki-Manome et al., 2020) on lake ice phenology and thickness.

Process-based models create significant benefit in their ability to provide sensitivity of ice season to climate variables and forecast ice cover conditions under the expected climate change. Modeling studies using climate projections predict ever-decreasing ice cover or no ice cover periods (Grant et al., 2021; Huang et al., 2022; Piccolroaz et al., 2021; Shatwell et al., 2019; X. Wang et al., 2022). Process-based models are also integrated into weather forecast and climate models to understand feedbacks between the atmosphere and lakes in current and future climate conditions (Mironov et al., 2010; Vavrus et al., 2015). Even short-term weather models have demonstrated the dependence of weather forecasts on accurate representation of lake ice conditions (Benjamin et al., 2022; Eerola et al., 2014; Fujisaki-Manome et al., 2020).

When data are limited, statistical models may struggle to learn and generalize relationships among variables. Similarly, process-based models may require simplifying assumptions that can impact their accuracy when scientific knowledge is incomplete. To address these limitations, a new category of models has recently emerged: hybrid models that integrate statistical methods with process-based approaches. Hybrid models make use of available data while maintaining a foundation in scientific principles, effectively combining multiple sources of information (Piccolroaz et al., 2024). Recently, machine learning methods have combined data-driven statistical models with process-based models into hybrid approaches to assess water quality (e.g., temperature and dissolved oxygen) and have been applied on a more limited basis to incorporate lake, river, and sea ice modeling (Andersson et al., 2021; L. Liu et al., 2022).

Despite advances in lake ice modeling, further advancements are necessary to enable the use of models for near-real time and seasonal ice cover forecasting. Near-real time ice cover forecasting directly affects the protection of life and property, and if integrated into other forecasting tools, it may be used by managers to inform safety conditions, mitigate winter drownings, aid in contaminant spill response, and support safe transportation and navigation. Seasonal ice cover forecasting can help managers plan for ice breakup and long-term ice-road transportation planning and anticipate risks to public ice access for recreation. Given the rapid advances of modeling and its potential for capturing fine-scale dynamics of lake ice cover, we created the following questions to highlight future avenues of research and existing challenges.

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## 7. What Are the Remaining Challenges and Big Questions for Modeling Lake Ice Research?

### 7.1. Forecasting of Ice Cover

44. How can forecasting tools for ice cover and duration be developed that capture intraseasonal dynamics?
45. How can 2D models improve our understanding of gradual ice formation that have been classified as open or frozen in 1D models?
46. Can medium-range (weekly to seasonal) lake ice forecasting meet the needs of society/communities?
47. How can lake ice modeling be used to inform scientists and lake managers about the influence of extreme events?
48. How well do models capture spatial and temporal variability in lake ice extent and thickness?
49. When should different process-based models (1D vs. 2D vs. 3D) be used for lake ice forecasting, and will this be impacted by climate change?
50. How can lake ice models be improved to accurately predict complex ice cover dynamics of large lakes ($>$100 km$^2$)?
51. How can lake ice models be improved to better predict ice cover dynamics during the melting period?
52. How does uncertainty in cloud cover, as a driver to ice dynamics, impact accuracy of lake ice models?
53. How can the changing porosity of ice be incorporated into current models to determine how safe ice cover is for human use?
54. Can available models determine multiple ice-on and ice-off events during the year, and how can those models be used to provide information on ice safety?
55. Can available models accurately capture littoral processes to better understand regions where less data exist?

Under-ice biogeochemistry

56. How can lake ice models capture biogeochemical fluxes and transformations to learn about the influence of nutrient fluxes and salts on lake ice conditions?
57. How reliable are models in predicting under-ice light conditions and to what level of confidence can specific processes influencing light attenuation, such as the presence of gas bubbles in the ice, be incorporated in lake ice models?
58. Can lake ice duration, thickness, and quality derived from models help contextualize important under-ice processes, nutrient fluxes or respiration, to understand ecological processes during the winter?

### 7.2. Improving Models

59. How can machine learning models be developed that can be applied at scale and through time with consistent accuracy, especially considering the diversity of inland water bodies?
60. Can hybrid models be used to capture ice thickness and ice quality during the freeze and melt periods when ice is too thin to sample, filling a gap where statistical models lack data?
61. What are the proper thresholds for identifying "complete" freeze-up timing in the ice models?
62. How can ice structure be incorporated into lake ice models to simulate developing ice cracks and improve safety forecasting?
63. What computational resources and model improvements are needed to increase temporal resolution of models, especially for three-dimensional models or those used for forecasting?
64. Does modeling ice cover on increasingly saline lakes require adaptation of existing models or entirely new models?
65. How can we improve simulations of snow on the ice surface, and what impact does that have on our estimates of ice thickness and ice composition?
66. What are the limitations of ice cover models in regard to "user requirements" and how can those limitations be overcome?
67. What is the optimal complexity regarding the number of modeled ice types for lake ice research questions?
68. Can 1D models be combined in a multi-column, quasi-2D or -3D model to improve computational efficiency and still retrieve necessary data to improve forecasting and answer ecological questions?
69. What is the best method to model bathymetric data when the data availability is limited and will modeled bathymetric data impact 1D, 2D, and 3D models similarly?

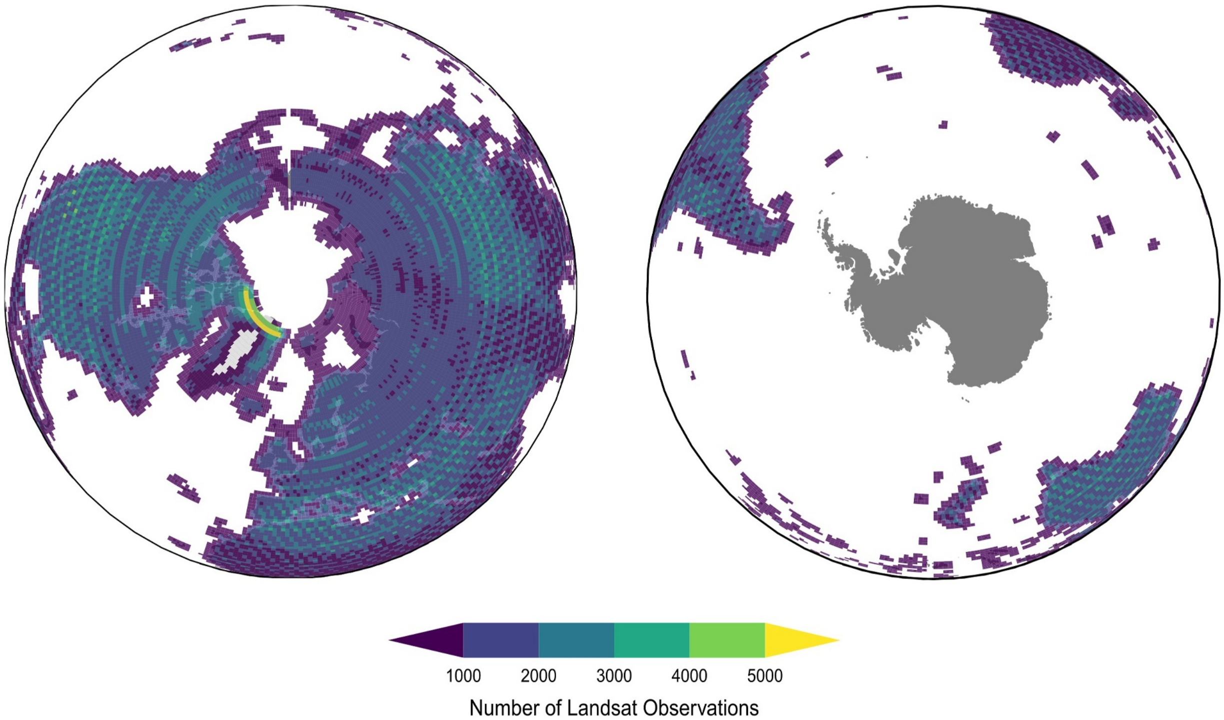


**Figure 3.** Global map of possible Landsat observations. The global coverage extent of the combined Landsat missions 5 through 9, except for Antarctica. The individual tiles represent a collected image and the coloring depicts the cumulative number of images for a given tile. The individual tiles have not been corrected for ascending and descending orbits or cloud cover. Landsat 5–9, Collection 2, Tier 1, provided by the U.S. Geological Survey (USGS) via Google Earth Engine (Gorelick et al., 2017).

70. Can lake ice modeling in arid regions be improved, given the sizable influence of sublimation and under-ice heating?

## 8. Opportunities for Integration Across Methods

Lake ice phenology has grown from collecting piecemeal lake ice records from around the globe to a concerted scientific effort to understand climatic and ecosystem consequences of lake ice loss (Benson et al., 2012; Magnuson et al., 2000; Sharma et al., 2022). Lake ice cover and lake ice thickness, derived from the aggregation and analysis of these records, are now recognized by the Global Climate Observing System (GCOS) as an Essential Climate Variable, being used to understand the state of the global climate by organizations such as the Intergovernmental Panel on Climate Change (Woolway et al., 2020). The rapid advances in understanding lake ice phenology over the last two decades through the methods articulated in the preceding sections beg the question: “What more can we learn by combining these resources?” This question can be especially important when we consider the unique information that each method provides (Figure 1). In situ observations have captured fine-scale, system-specific changes in lake ice loss that both pre-date remote sensing image records (Magnuson et al., 2000) and contain information without generalizable analytical and empirical relationships (Sharma et al., 2022). Remote sensing technologies have allowed for standardized, global-scale analyses of the spatial and temporal distributions of lake ice, thereby enabling investigations of data-sparse or logistically difficult locations (Q. Yang et al., 2023) yet still have data gaps and a northern-centric data focus (Figure 3). Lake ice modeling techniques have leveraged computational efficiencies to operate at finer time and space scales than either in situ monitoring or remote sensing. The dynamics of ice phenology can be estimated across ecosystems when only a handful of meteorological, climatological, and lake morphometric variables may be known a priori (Stefan, 1891; Weyhenmeyer et al., 2011).

But what are the benefits of integrating each of these perspectives to understand lake ice cover? Integrating the three methods of inquiry highlighted above engages with the limitations of aggregating large-scale data collection techniques (i.e., ecoinformatics), strengthens study design by broadening the scope of possible questions, and

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combines interdisciplinary knowledge that borders hydrology, climatology, and ecology (Sharma et al., 2020). Below, we detail how each of these interdisciplinary areas may be substantially progressed by integrating the study of lake ice phenology across in situ, remote sensing, and modeling fields.

## 9. Integration Across Methodologies Promotes Standardization of Ice Phenology Data

One of the most immediate differences across in situ, modeling, and remote sensing techniques is how each field classifies "ice-on" and "ice-off." In situ studies may record phenology in terms of binary instances, where "ice-on" and "ice-off" may be associated with particular dates as well as the viewpoint of a given monitoring location (Williams et al., 2004). Even within and across in situ studies, defining "ice-on" may be problematic (Magnuson et al., 2000; Sharma et al., 2022). Smaller lakes may freeze completely within a few days, and the freezing completion may not always be visibly apparent. In large lakes, such as the great lakes of the world (i.e., >500 km$^2$), ice does not freeze on the whole lake instantaneously, meaning that localized assessments of ice phenology may not be representative of the entire system (Ozersky et al., 2021; J. Wang et al., 2018). Modeled ice phenology data may arise from simplified assessments of ice formation, and because many tractable models may calculate ice development in one dimension, they can offer a limited viewpoint of ice phenology but with the added benefit of interpolating information about lake ice between moments with actual observations (Croley & Assel, 1994; Vavrus et al., 1996). Remote sensing ice phenology estimates allow for the most disaggregated understanding of lake ice; where rather than a binary or continuous estimate of ice at a point in space, remote sensing offers the capacity to think of ice presence in terms of continuous areas through space and time (Brown & Duguay, 2010; Q. Yang et al., 2023).

True integration across in situ, remote sensing, and modeling efforts must reckon with how information of lake ice phenology is recorded. Each technique comes with its own tradeoffs, yet there is potential for each technique's acquired data to have limited interoperability (see Figure 1 in Sharma et al., 2020). A major challenge for understanding lake ice cover will be how to best operationally and synthetically combine binary, interpolated, and continuous data. Looking to the future, integration of methods necessitates understanding and standardizing how data are collected, thereby allowing each technique to take full advantage of the rich data troves that each technique provides. The following questions center on the best methods to unify data sources.

71. How do lake ice researchers create standards for data collection and data sharing?
72. How can lake ice researchers calibrate and validate data from in situ sampling, models, remote sensing, and new, emerging methodologies?
73. To what extent does reservoir ice behave universally, when reservoirs are used for a multitude of societal needs but have fundamental differences from naturally forming lakes?
74. Is the integration of ice monitoring techniques amenable to both natural lakes and reservoirs?
75. Can a common metadata ontology be derived across geographic regions by classifying lakes by their characteristic ice phenology regimes?
76. Do traditional lake classification approaches, such as trophic state, lake districts, and hydrologic landscape connectivity, capture potential increases in spatial heterogeneity for intermittent ice lakes?
77. How can error and variability between data collection methods be accounted for to inform inferences from records created by hybrid sources?
78. Can a subset of lakes act as model lakes and case studies for validation between techniques?
79. What is the data cost of having a larger number of lakes monitored with a limited resolution (temporal or spatial) versus having a small number of lakes or lake representatives well monitored? Does data quality outrank data quantity when asking landscape-scale questions?
80. How can lake ice researchers integrate local knowledge and Traditional Ecological Knowledge into data collection and lake ice phenological analyses?

## 10. Integration Across Methodologies Can Further Understanding at Multiple Spatial and Temporal Scales

A common theme in the environmental sciences is how patterns and processes are observable across disparate spatial and temporal scales (Soranno et al., 1999). Lake ice is no exception. Magnuson et al. (1990) is a classic example of how limited observation of in situ lake ice onset across years can obfuscate clear declines in lake ice duration, where temporal patterns are most apparent at the decadal-centurial timescale. Spatially, study of lake ice phenology tends to occur in the Northern Hemisphere, yet lake ice phenology in the Southern Hemisphere may reveal alternative climatic connections (Huang et al., 2022). The spatial and temporal scales at which we understand lake ice phenology ultimately stems from the methodological approaches employed. Where available, in situ observations can inform lake ice phenology across centuries but at limited spatial scales (Sharma et al., 2022). The ever-increasing array of satellites with sensors aimed at surface waters increases the observational capacity of limnologists with an eye on global-scale lake processes (Carrea et al., 2023; Fu et al., 2023; Giroux-Bougard et al., 2023). Lastly, modeling techniques permit inferences of lake ice phenology at fine scales within individual ecosystems worldwide, owing to increased availability of high-frequency meteorological data and high-performance computing.

True integration across in situ, remote sensing, and modeling methodologies must accept that insights afforded by one methodology may not be compatible across spatial and temporal scales (Sharma et al., 2020). In situ efforts are ultimately constrained by the physical and logistical capacities associated with site-based research. Much like integrating data, each technique comes with scale tradeoffs. Coordinated in situ sampling may give synoptic interpretations of how lake ice phenology varies across a landscape (Magnuson et al., 2000), but field-based efforts will never give the completeness or detail of modeling and remote sensing techniques. Therefore, a major challenge for lake ice study will be synthesizing how insights made at a certain spatial or temporal scale may or may not be reproducible at other scales. Looking to the future, lake ice cover research can engage in the larger discussion about how scale governs environmental patterns and processes, thereby expanding the spatial and temporal domains that pertain to lake ice research. Therefore, we have gathered the following questions that engage with the central premise of limitation and capacity across methodologies.

81. At what temporal or spatial scale do individual techniques no longer capture ice phenology accurately?
82. How do complex socio-ecological systems within and across scales influence lake ice phenology?
83. How temporally consistent is spatial variability in ice thickness?
84. What is the spatial synchrony of ice thickness within a lake and between nearby lakes?
85. How does lake ice loss affect the global water cycle?
86. Does a lake's size or location impact its applicability as a sentinel of climate change regarding future lake ice changes?
87. To what degree can increasing evaporation rates caused by declining ice cover duration cause a positive feedback loop, and are there common patterns across large spatial scales?
88. How do urban environments that experience heat island effects influence ice phenology?
89. Can winter observations be harmonized with open-water periods to more cohesively describe seasonal limnology?

## 11. Integration Across Methodologies Can Increase the Interdisciplinarity of Lake Ice Study

Lake ice phenology is an interdisciplinary field (Block et al., 2019; Sharma et al., 2020); however, methodologies that vary between individual studies to understand patterns in lake ice phenology run the risk of siloing lake ice studies from established concepts in other fields. From a hydrological perspective, lake ice is a critical part of the global water cycle (Downing, 2010; Pi et al., 2022), but the extent to which lake ice loss influences water budgets across landscapes and lake chains is uncertain and might be context dependent. From a climatological perspective, decreasing lake ice is the manifestation of rising air (Magnuson et al., 2000; Sharma et al., 2016) and water temperatures (O'Reilly et al., 2015), where physical properties of ice and water may be consequential for heat budgets. From an ecological perspective, lake ice phenology can be thought of as an ecological disturbance (Bramburger et al., 2022; Hampton, Galloway, et al., 2017), where uninterrupted ice cover can be viewed as a press disturbance, whereas freeze-thaw events can function as pulse disturbances (Jentsch & White, 2019). However, the full extent of the impact of changing lake ice cover on under-ice ecology remains unclear (Culpepper et al., 2024; Hampton, Galloway, et al., 2017).

In each of these disciplinary offshoots—integrating in situ, remote sensing, and modeling—efforts can increase winter limnology's capacity to engage other disciplines and to reorient winter limnology as a seasonal component of limnological studies, rather than a distinct discipline. True integration of these methods would allow for a more nuanced understanding of lake ice formation and thaw (Sharma et al., 2020), thereby empowering a suite of disciplines to assess how lake ice loss may influence their research objectives. Operationally, engaging other

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fields will require the data and concepts related to winter limnological research to be transferable to limnology more broadly (i.e., open-water methodology) and to other disciplines to facilitate the use of winter limnological data and methods to explore questions and undergird research in other fields. Furthermore, this investment in the interdisciplinarity of winter limnology can encourage growth and versatility in the data collection methods described above. Interdisciplinary collaborations may necessitate alternative integrations of in situ, modeling, and remote sensing, thereby enhancing how we can use information from methodological integrations to produce the most cutting edge science (Sharma et al., 2020).

90. What metadata standards should be implemented to maximize interdisciplinary reuse of lake ice phenology and quality data?
91. What role does changing lake ice phenology play in the global water cycle?
92. How do multiple freeze-thaw events contribute to nutrient cycles?
93. How does ice phenology influence the transport and accumulation of emerging contaminants of concern, such as pharmaceutical and personal care products, volatile organic compounds, as well as perfluoroalkyl and polyfluoroalkyl substances?
94. How do shifts in lake ice phenology influence geopolitical relations and how do shifts in geopolitical relations influence data sharing and interoperability?
95. How does lake ice loss impact behavior and survivorship of migratory terrestrial and aquatic species?
96. How does lake ice loss influence life stages of ice-obligate taxa?
97. Can winter limnology techniques improve the understanding of dangerous hydrological phenomena such as potentially increasing lake ice outburst floods as ice dams decrease?
98. What impact (e.g., changing global surface albedo or energy budgets, etc.) does lake ice phenology have compared to other components of the cryosphere, such as glaciers and sea ice?
99. Can time-varying ice freeze-thaw events be viewed as press-pulse disturbances?

## 12. 100. How Do We Pragmatically Move Forward?

Given the promise of integrating in situ, remote sensing, and modeling efforts to understand shifting lake ice phenology, pragmatically moving forward with integrating these methods will require building a workforce savvy in these techniques and the aquatic sciences. Currently, training in computationally intensive environmental science as well as informatics is highly variable across institutions (Hampton, Jones, et al., 2017; Sharma et al., 2020). Beyond training, discipline-specific data storage conventions and formats can vary in structure and complexity across fields, further siloing lake ice study (Hampton, Jones, et al., 2017; Sharma et al., 2020).

Therefore, the future lake ice phenology workforce would likely benefit from a multi-tiered effort to scale the number of skilled researchers with the pace of a changing climate. Program leads may need to invest resources into supporting grassroots efforts, such as the creation of working groups that brainstorm synergies or communities of practice focused on skill development and knowledge sharing. The Global Lake Ecological Observatory Network's (Weathers et al., 2013) Winter Limnology Working Group or the National Science Foundation Lake Ice Workshops could be examples for how to engage winter-focused freshwater scientists in an environment that encourages multidisciplinary collaboration. With a focus on the nexus of remote sensing, data science, open science, and the aquatic sciences, the Community for Data Science and Open Science (Meyer et al., 2021, 2022; Meyer & Zwart, 2020) could be a base for skill development, where lake ice enthusiasts gain intermediate level skills in cutting edge computational techniques. Through investment in grassroot initiatives, lake ice phenology may become an even more widespread phrase associated with a changing climate, where even the beginning student appreciates and can engage in integrating in situ, remote sensing, and modeling lake ice observations as part of their standard training.

## Data Availability Statement

Satellite imagery for Sentinel-1 and Sentinel-2 are freely available via the European Space Agency at https://documentation.dataspace.copernicus.eu/Data/SentinelMissions/Sentinel1.html and https://documentation.dataspace.copernicus.eu/Data/SentinelMissions/Sentinel2.html. Landsat data are freely available via the U.S. Geological Survey (USGS) at https://www.usgs.gov/landsat-missions/landsat-data-access.

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